

Battery voltage divider for measuring voltage

R4 = 470
R5 = 680

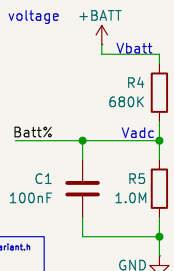
Voltage Division Equation
Vadc = Vbatt * (R5 / (R4 + R5))

Voltage Vbatt --> Vadc ratio
ADC_Division = Vadc / Vbatt = (R5 / (R4 + R5))
R5 / (R4 + R5) = 0.591304347826087

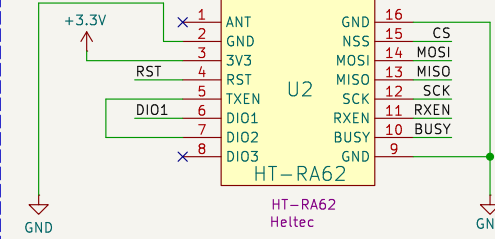
Voltage Vadc --> Vbatt ratio
ADC_Multiplier = Vadc / Vbatt = (R4 + R5) / R5
(R4 + R5) / R5 = 1.6913764705882353

C1 used for getting more stable and accurate readings

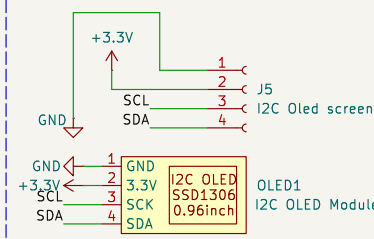
firmware/variants/nrf52840/diy/nrf52_promicro_diy_txco/variant.h
#define VBAT_DIVIDER_COMP (1.73)
#define ADC_MULTIPLIER VBAT_DIVIDER_COMP



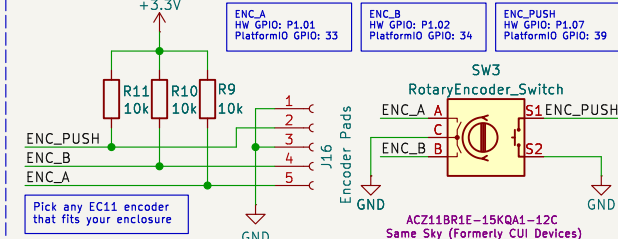
HT-RA62 / RA-01SH



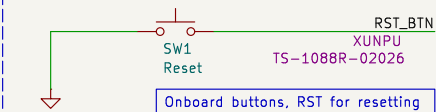
Pins for connecting an I2C OLED screen



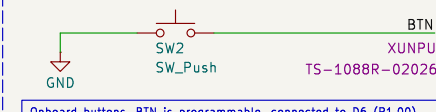
Pads for connecting encoder



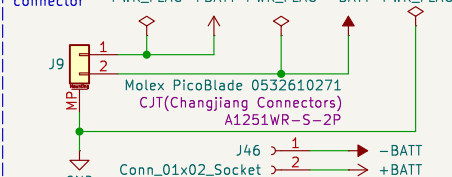
Onboard buttons RST



Onboard buttons BTN



Battery connector



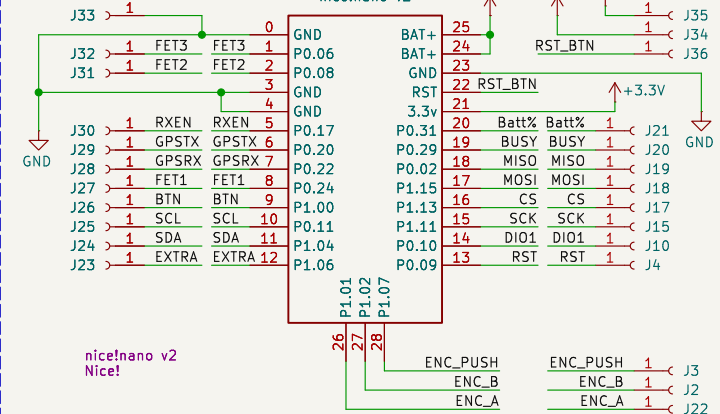
Molex PicoBlade 2-Pin
- MPC: Molex, MPN: 0532610271
- MPC: CJT (Changjiang Connectors), MPN: A1251WR-S-2P
- Pitch: 1.25mm
- Used By: HelTec LoRa 32
- This polarity is also common for lipos in Aliexpress

JST PH 2-Pin
- MPN:S2B-PH-SM4-TB
- Pitch: 2.00mm
- Used By: RAK19007 WisBlock
- This polarity is also common for lipos in Aliexpress

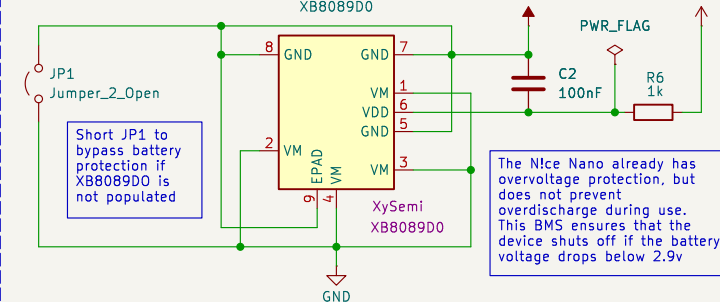
EXTRA GPIO



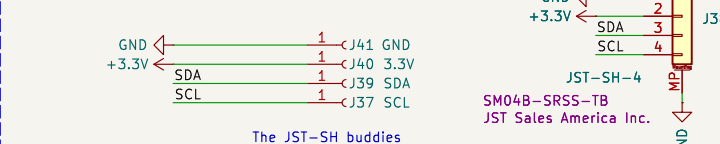
Nice!Nano nrf52840 microcontroller (or promicro)



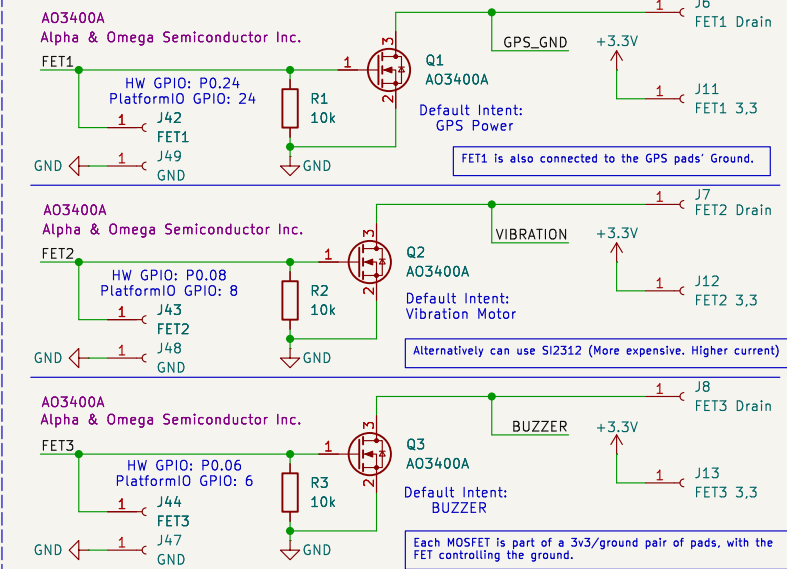
BMS for overdischarge protection



I2C Port (For extra sensors etc...)



Mosfets used for switching high current external perpherial



GPS Module
+3.3V
GPS_GND
GPSRX
GPSTX
e.g. Buy either ATGM336H or GY-GPS6MV2 modules

Vibration Motor
+3.3V
VIBRATION
e.g. Can buy in bulk very cheaply from other sources

Buzzer
+3.3V
BUZZER
e.g. Can buy in bulk very cheaply from other sources